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RESEARCH-ARTICLE

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The Influences of a Knowledge Representation Tool on Searchers with Varying Cognitive Abilities

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While current systems are effective in helping searchers resolve simple information needs (e.g., fact-finding), they provide less support for searchers working on complex information-seeking tasks. Complex search tasks involve a wide range of (meta)cognitive activities, including goal-setting, organizing information, drawing inferences, monitoring progress, and revising mental models and search strategies. We report on a lab study ($N = 32$) that investigated the influences of a *knowledge representation* tool called the OrgBox, developed to support searchers with complex tasks. The OrgBox tool was integrated into a custom-built search system and allowed study participants to drag-and-drop textual passages into the tool, organize passages into logical groupings called “boxes”, and make notes on passages and boxes. The OrgBox was compared to a baseline tool (called the Bookmark) that allowed participants to save textual passages, but not organize them nor make notes. Knowledge representation tools such as the OrgBox may provide special benefits for users with different cognitive profiles. In this article, we explore two cognitive abilities: (1) working memory (WM) capacity and (2) switching (SW) ability. Participants in the study were asked to gather information on a complex subject and produce an outline for a hypothetical research article. We investigate the influences of the tool (OrgBox vs. Bookmark) and the participant’s working memory capacity and switching ability on three types of outcomes: (RQ1) search behaviors, (RQ2) post-task perceptions, and (RQ3) the quality of outlines produces by participants.

CCS Concepts: • **Information systems** → **Users and interactive retrieval**;

Additional Key Words and Phrases: Knowledge representation, working memory capacity, switching ability, complex search

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1 INTRODUCTION

Current search systems are effective in helping users with simple search tasks (e.g., fact-finding). However, they provide less support for users during complex search tasks that require a diverse set of (meta)cognitive activities, such as setting goals, monitoring progress, making inferences, updating mental models, and revising strategies for completing the task. One approach to supporting searchers with complex tasks have been to develop *search assistance* tools that are *complementary* to the search interface. Prior research has investigated a wide range of tools to help searchers save, organize, and annotate documents as they work on a complex search task [4, 33, 35, 36, 57, 79].

In this article, we present research on a *knowledge representation* tool called the “OrgBox” (Figure 3). We report on a laboratory study that investigated the influences of the OrgBox tool (compared to a baseline tool) during complex search tasks. We specifically designed the OrgBox tool to support the types of (meta)cognitive activities involved during complex search tasks. The OrgBox was integrated into a custom-built search system and was designed to help searchers easily save and organize information by drag-and-dropping textual passages from a landing page onto logical groupings called “boxes”. Using the OrgBox, study participants could add, delete, and edit passages and boxes; add notes to passages and boxes; indent passages within a box to create hierarchies; move passages between boxes; and reposition boxes around the interface. These features were designed to support users with developing subgoals, saving and (re-)organizing information, representing relationships, and monitoring their progress. To study the influences of the OrgBox tool, we compared it against a baseline tool called the “Bookmark” (Figure 4). The Bookmark tool allowed participants to drag-and-drop textual passages onto a vertical list but not (re-)group them, (re-)arrange them, nor make notes.

We refer to the OrgBox as a knowledge representation tool because it enables searchers to *externalize* and *modify* their knowledge structures as they gather information and learn about a complex topic. While there are numerous theories of how knowledge is represented in the human mind, fundamental components include facts, concepts, and relations [26, 30, 66]. Features of the OrgBox allowed participants to represent important topics and relations among them. Additionally, prior work suggests that external knowledge representations can promote and support important (meta)cognitive activities, including setting goals, monitoring progress towards specific goals, and revising goals and strategies [78].

Prior research has investigated how user characteristics can influence search behaviors and outcomes. For example, studies have considered *personality traits* such as the need for cognition [80] and *cognitive abilities* such as perceptual speed [9], **working memory (WM)** capacity [3, 31, 32], and attention control [3]. However, studies have not investigated how specific cognitive abilities may influence how individuals experience the benefits (or challenges) introduced by tools such as the OrgBox, designed to support (meta)cognitive activities required by complex search tasks.

In this article, we consider the impact of two cognitive abilities—WM capacity and **switching (SW)** ability.

WM capacity relates to an individual’s ability to hold and manipulate information in short-term memory. WM capacity has been found to influence performance on tasks such as vocabulary acquisition [21], language comprehension [22], and arithmetic problem-solving [1]. Plenty of research in psychology has found a connection between WM capacity and cognitive control (see Oberauer [59] for review). The exact nature of this connection remains an area of active research. For example, one perspective argues that WM contributes to controlling *perceptual attention* and controlling *action* [59]. In terms of controlling perceptual attention, WM allows individuals to hold characteristics of perceptual targets in active memory. In terms of controlling action, WM enables individuals to hold and manipulate action plans in active memory. Studies have found that

WM capacity influences an individual's ability to suppress interference (e.g., non-relevant information) [43, 45]. Studies have also found that WM plays a critical role in supporting top-down, goal-oriented behaviors [17, 28, 38].

SW ability relates to an individual's ability to easily and effectively shift between different types of activities or perspectives. To illustrate, Rourke et al. [61] found that children with high SW ability generated more reasonable action plans when faced with changes in the task requirements. Similarly, research has found an association between SW ability and arithmetic problem-solving [10, 14, 52, 60, 76]. The intuition is that arithmetic problem-solving requires switching between arithmetic fact-finding and calculation strategies.

An important aim of our research was to study the effects of WM capacity and SW ability during complex search tasks assisted by tools such as the OrgBox or Bookmark. To our knowledge, prior research has not considered the influences of these two cognitive abilities in this context. Our decision to focus on WM and SW is motivated by several trends observed in prior work.

Working Memory: The OrgBox was designed to help searchers *externalize* their knowledge structures while gathering information (e.g., create/rearrange topical clusters and make notes). On one hand, tools such as the OrgBox may provide benefits for users with deficits in WM capacity, by allowing them to create *external* representations of their ideas and action plans and free-up mental resources for other cognitive tasks. On the other hand, prior research has also found that high-WM individuals tend to exert more effort in challenging situations [13, 31, 32] and exhibit more top-down attentional control [6, 15, 17, 43, 49, 75]. Thus, in our case, higher-WM participants may be more willing and able to leverage the additional complexity of the OrgBox (vs. Bookmark) tool to achieve better outcomes.

Switching Ability: Compared to the Bookmark tool, the OrgBox requires searchers to shift between different types of cognitive activities (e.g., searching for information, creating boxes, rearranging boxes). In this respect, searchers with higher SW ability might be more willing and able to engage with the OrgBox and achieve better outcomes.

We report on a laboratory study ($N = 32$) in which participants completed two complex search tasks, one with the OrgBox and one with the Bookmark tool. Our search tasks were situated in the following scenario. Participants were asked to imagine being a journalism student with an assignment to write a 30-page article on a given complex topic in U.S. politics (abortion or gun control). During each task, participants were asked to gather information using a custom-built search engine and the assigned tool. During the *search phase* of each task, participants were given 40 minutes to gather information. Then, during the *outline generation phase* of each task, participants were given 15 minutes to produce an outline for the hypothetical 30-page article.

Our study investigated three main research questions (RQ1–RQ3). Each research question focused on the effects of the assistance tool (i.e., OrgBox versus Bookmark), the participant's WM capacity, and the participant's SW ability on: (RQ1) search behaviors, (RQ2) post-task perceptions, and (RQ3) the quality of the outline produced.

To investigate RQ1, we logged all user interactions with the system (i.e., search engine + assistance tool). Using this logged data, we computed behavioral measures associated with the participant's querying behavior, the pace of interaction, search effort, and the extent to which the participant adopted search strategies that *diverged* from other participants.

To investigate RQ2, after the search and outline generation phase of each task, participants completed a post-task questionnaire that was divided into three sections. The first section asked questions about the participant's experience during the task (e.g., difficulty, interest, satisfaction). The second section asked questions about the quality of the outline produced by the participant along different dimensions (e.g., breadth, depth, logical flow). The third section asked about the

extent to which the assigned tool (i.e., OrgBox or Bookmark) helped with different (meta)cognitive activities (e.g., planning, monitoring progress, revising strategies).

To investigate RQ3, we performed a qualitative analysis of the outlines produced by participants. Outlines were rated along six dimensions: (1) breadth, (2) depth, (3) a balance of breadth and depth, and (4–6) three dimensions associated with the extent to which the outline had a logical organization of topics and subtopics.

Our study makes four important contributions. First, it adds to the literature on the potential benefits (and challenges) introduced by tools that help searchers externalize their knowledge structures as they search. Second, it extends our understanding of the role of WM during complex search tasks. Third, we are the first to consider the effects of SW ability on search behaviors and outcomes. Finally, we consider the potential benefits (and challenges) introduced by knowledge representation tools for searchers with varying levels of WM capacity and SW ability.

2 RELATED WORK

Our study builds on prior work on knowledge representation tools, as well as prior work on the effects of WM capacity and SW ability during information-seeking (and related) tasks.

2.1 Knowledge Representation Tools

Research in cognitive psychology argues that externalizing knowledge structures can help individuals store, structure, and utilize knowledge [39, 58]. Additionally, creating visual knowledge representations can help individuals create more coherent representations of *new* information [39]. In education, it is argued that externalizing knowledge can help students organize, analyze, evaluate, and generate ideas [29, 42, 71]. Bransford et al. [7] summarize several studies that found that learners who categorized information were more likely to recall it. Tergan [71, 72] argues that computer-based concept mapping tools can help learners: (1) organize information, (2) maintain awareness of knowledge gaps, (3) evaluate the quantity/quality of their own knowledge, (4) generate new knowledge, and (5) communicate knowledge to others.

More closely related to IIR, the role of knowledge structures has been studied in the context of *sense-making*, which involves “making sense” of a complex domain or task. Russell et al. [62] originally conceptualized sense-making as a cyclical process. The first stage involves searching for a good representation (i.e., the “generation loop”). The second stage involves finding and encoding information using the latest representation (i.e., the “data coverage loop”). The data coverage loop often involves identifying information that does not fit neatly into the latest representation (i.e., the “residue”), which requires adjusting the latest representation to have better coverage (i.e., the “representation shift loop”). Russell et al. [62] called for a tighter integration between IR systems and knowledge representation tools. Similarly, Stefik et al. [70] argued that when a sense-making task is difficult, sense-makers need *external* representations to store information for repeated manipulation.

Crescenzi et al. [20] conducted a lab study to learn about the design requirements for knowledge representation tools that support users with complex searches. The study involved a qualitative analysis of handwritten notes made by participants during exploratory search tasks. Results found that participants used different structures to organize information (e.g., single-level lists, multi-level lists, and clusters). Additionally, simple structures (e.g., lists) were more common at earlier stages, while complex structures (e.g., clusters) typically emerged as the search progressed.

Prior research has developed and evaluated a wide range of tools to help users save and organize search results. Hearst and Degler [35] developed a tool similar to our OrgBox tool. The tool allowed participants to save, delete, visually group, and annotate documents. A usability test with

legal analysts found that participants responded favorably to the tool. Bae et al. [4] developed the **Visual Knowledge Building (VKB)** system, which allowed users to group and annotate documents. However, the tool was not integrated with a search system. Results from a usability study found that some participants preferred to organize documents as they evaluated their relevance, while others preferred doing document triage first and organization second. Hinckley et al. [36] designed the InkSeine system, which allowed users to organize search results into “piles” and add notes. The Sandbox system was developed to help intelligence analysts with complex information-seeking tasks [79]. The tool allowed users to save, group, and annotate search results. Results from a small usability study found that the tool allowed participants to examine more documents in less time. In the context of multimedia search, prior work developed the ViGOR system for video search [33] and ImageGrouper for image search [57]. Both systems allowed users to save search results into groups and use groups as queries. (i.e., querying by groups of examples).

To summarize, many tools like the OrgBox have been designed to help searchers organize information. Our research builds on this prior work by investigating whether such tools provide specific benefits (or introduce challenges) for searchers based on their WM capacity and SW ability.

2.2 Working Memory

WM capacity is defined as someone’s ability to hold information in short-term memory when it is no longer perceptually present. WM is considered a core executive function and is critical for making sense of anything that unfolds over time. In this respect, WM capacity impacts higher-level tasks such as reading, following instructions, incorporating new information into action plans, and generalizing from particular observations. WM capacity has been found to predict performance in vocabulary acquisition [21], language learning [22], arithmetic problem-solving [1], following directions [25], and note-taking [46].

The *controlled attention* perspective of WM suggests that WM also influences an individual’s ability to maintain top-down attentional control and remain focused while suppressing interference [43, 45]. Researchers have found that high-WM participants are more likely to exhibit goal-oriented and purposeful attention control [6, 15, 17, 43, 49, 75]. In a survey by Wiley and Jarosz [77], the authors argue that one of the main ways in which WM benefits problem-solvers is by limiting distraction from irrelevant problem features. Kane et al. [44] conducted a study in which participants were polled 8 times a day for a week and asked about their current activity and whether their thoughts had wandered from the task. During challenging tasks that required concentration, high-WM participants reported less mind-wandering than low-WM participants. Sanchez and Wiley [64] conducted a study in which participants were asked to read passages on a given topic and write an essay response. Passages were presented in three conditions: (1) text-only, (2) text with topically relevant images, or (3) text with topically irrelevant images. High-WM participants produced essays of similar quality in all three conditions. Conversely, low-WM participants produced essays of lower quality in the irrelevant images condition.

Studies have also investigated the role of WM on learning within different environments. Sanchez and Wiley [65] conducted a study in which participants read passages and wrote an essay response. In one condition, passages were displayed on different webpages. In the other condition, passages were displayed in one long webpage that required scrolling. High-WM participants produced essays of similar quality in both conditions. Conversely, low-WM participants produced lower quality essays with the scrolling interface. The authors hypothesized that presenting passages in different webpages encouraged participants to engage in “wrap-up” processes that are helpful for reading comprehension (i.e., internalizing the current passage before moving to the next). The authors argued that participants neglected these “wrap-up” processes with the scrolling interface and that this had a more negative effect on reading comprehension for low-WM

participants. In a study by Banas and Sanchez [5], participants were asked to study a set of web-pages that were organized in a topical hierarchy, from general concepts to specific ones. High-WM participants outperformed low-WM participants in a “tree construction” task that required them to illustrate hierarchical relations between concepts.

Several studies have investigated the role of WM in information seeking. Gwizdka [31] investigated the interplay between WM, search task complexity, and two different interfaces—a baseline search system and an experimental system that displayed a word cloud alongside the search results. In more demanding situations (i.e., during complex tasks with the baseline system), high-WM participants exerted more search effort. In another study, Gwizdka [32] used eye-tracking to compare the search activities of low- and high-WM participants. High-WM participants spent more time reading pages, particularly towards the end of the session. Choi et al. [13] investigated the role of WM during search tasks of varying complexity. Participants completed decision tasks that required them to compare alternatives, make a selection, and write a justification. Tasks complexity and WM had an interaction effect on the length of justifications written by participants. During simple tasks, high- and low-WM participants produced justifications of similar length. Conversely, during complex tasks, high-WM participants produced longer justifications. Arguello and Choi [3] investigated the role of WM on participants’ post-task perceptions of workload and engagement. Participants interacted with two aggregated search interfaces—the *blocked* interface organized the search results by vertical (web, images, news, shopping, video) and the *interleaved* interface was more visually cluttered. WM and the interface condition had a significant interaction effect on participants’ perception of mental demand, which was measured using the NASA-TLX [34] questionnaire. While low-WM participants reported similar levels of mental demand with both interfaces, high-WM participants reported lower mental demand with the blocked interface. The authors hypothesized that high-WM participants were better able to recognize and memorize the fixed layout of the blocked interface, allowing them to focus their attention on relevant verticals.

Prior studies have also investigated the search behaviors of dyslexic users. While dyslexia is a heterogeneous condition, it is associated with deficits in verbal and visuospatial WM [69, 73]. In a study by MacFarlane et al. [50], non-dyslexic participants viewed more documents and judged a greater percentage of documents as being non-relevant. In other words, non-dyslexic participants were more “selective”. MacFarlane et al. [51] further examined the impact of dyslexia using eye-tracking. Results found that dyslexic participants were more likely to revisit previously viewed information. Morris et al. [56] studied the challenges faced by dyslexic searchers through in-depth interviews, an online survey, and a lab study. Based on the interviews and survey responses, dyslexic searchers reported preferring content with less clutter (e.g., less ads), less density of text (e.g., shorter sentences), and more structural elements (e.g., lists, tables). In the lab study, participants were asked to rate webpages along dimensions of understandability. Ratings from dyslexic participants were less positive. Fourney et al. [27] compared relevance judgments produced by dyslexic versus non-dyslexic participants. Non-dyslexic participants produced relevance judgments that were more bimodal (i.e., either highly relevant or non-relevant). Conversely, dyslexic participants produced relevance judgments with a greater central tendency, suggesting they had more difficulty distinguishing between relevant and non-relevant pages.

To summarize, prior studies have investigated the impact of WM on different search behaviors and outcomes (e.g., post-task perceptions and objective measures of performance). However, as mentioned above, prior studies have not investigated the potential benefits or challenges introduced by knowledge representation tools such as the OrgBox for searchers with varying levels of WM capacity.

2.3 Switching Ability

SW ability is defined as someone's ability to shift attention between tasks, perspectives, or strategies, and is related to cognitive flexibility [55, 81]. It is also defined as an individual's ability to adapt behaviors in response to changes in internal goals and/or the external environment [24, 54]. Switching is considered one of the core **executive functions (EF)**, which are required to perform higher-order thinking. According to Diamond [23, 24], there are three core EFs: WM, inhibition, and cognitive flexibility (i.e., switching). In turn, these three core EFs support higher-order EFs such as critical thinking, problem-solving, and reasoning. SW ability is important because many cognitive tasks require information processing to be controlled and reallocated across different components [63]. SW ability allows individuals to change their attention from one stimulus, representation, or process to another according to internal goals and task demands.

Studies have found that people differ in their ability to easily and effectively shift from one task to another. Salthouse et al. [63] investigated the relation between SW ability, higher-order cognition, and age. Results found that older participants had more difficulty SW, and participants with greater SW ability scored higher on other measures of cognition (i.e., inductive reasoning, spatial visualization, and episodic long-term memory). Studies have also found associations between SW ability and performance on tasks such as reasoning, reading, and mathematical problem-solving. Rourke [61] found that children with higher SW ability generated more reasonable action plans when faced with changes in the task requirement. Colè et al. [16] found that SW ability predicted reading skills in second graders. Additionally, plenty of research has found an association between SW ability and arithmetic problem-solving [10, 14, 52, 60, 76]. The intuition is that arithmetic problem-solving requires shifting between arithmetic fact-finding strategies and calculation strategies.

To our knowledge, prior work has not investigated the role of SW ability during complex search tasks. However, prior work has considered the role of SW ability in hyperlinked environments. Scharinger et al. [67] conducted a study in which participants were asked to read text that was displayed in either plain text or hypertext. Participants exhibited greater cognitive load (e.g., pupil dilation) in the hypertext condition. The authors hypothesized that the hypertext environment forced participants to *switch* between different modes of working: reading and evaluating/selecting hyperlinks.

3 METHODS

3.1 Study Overview

To address RQ1–RQ3, we conducted a laboratory study with 32 participants (26 female). Participants were recruited using an opt-in mailing list of employees from our university. Participants included 17 student employees and 15 non-student employees. Their ages ranged from 18–66 ($M = 29.16$, $S.D. = 11.73$). Participants had a wide range of educational backgrounds. In terms of highest degree or level of schooling completed, two participants reported having some college credits, eight had a bachelor's degree, five had a graduate degree, and 17 did not specify. Similarly, 16 participants reported having a STEM-related degree, six had degrees in the social sciences, three had degrees in the arts, and seven did not specify.

Participants completed the study in two separate sessions, which took place on different days. The study was conducted in the Fall of 2019, prior to the COVID-19 pandemic. Thus, both study sessions were conducted in-person in our lab space at UNC. During the first session (day one), participants completed two experimental tasks. Each task proceeded in two phases. During the *search phase*, participants were asked to use a custom-built search system to gather information on a specific multifaceted topic in U.S. politics (i.e., either abortion or gun control). During the

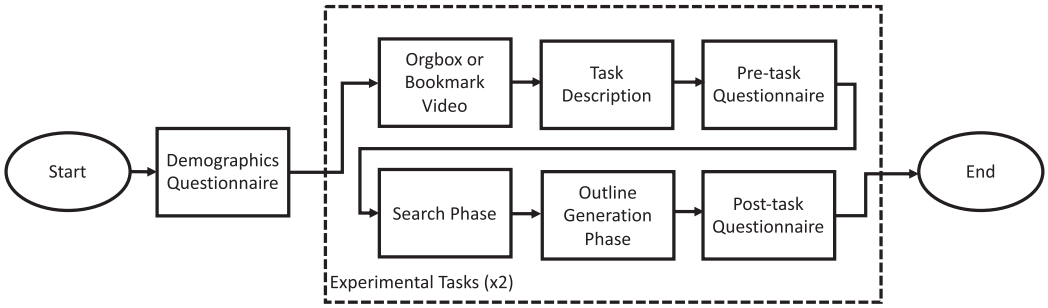


Fig. 1. First session protocol.

outline generation phase, participants were asked to generate an outline of a hypothetical article (comprehensive and well-balanced) on the assigned topic. During the second study session (day two), participants completed two psychometric tests that measured WM capacity and SW ability. Participants completed the psychometric tests on a different day as the main study session to help prevent fatigue effects. The first session of the study took about 2 hours and the second session took about 30 minutes. Participants were given U.S. \$40 for participating.

During the first study session, participants were exposed to two tasks (i.e., abortion and gun control) and two tools to organize the information found (i.e., OrgBox and Bookmark). This resulted in four possible treatment orders: (1) abortion/orgbox → gun/bookmark; (2) gun/bookmark → abortion/orgbox; (3) abortion/bookmark → gun/orgbox; and (4) gun/orgbox → abortion/bookmark. An equal number of participants experienced each treatment order (i.e., $4 \times 8 = 32$) and each task-tool combination (i.e., $2 \times 16 = 32$).

3.1.1 First Session Protocol. The protocol of the first study session is illustrated in Figure 1. After getting an overview of the study and providing informed consent, participants completed a demographics questionnaire. Then participants completed two experimental tasks, which followed the same sequence of steps. First, participants were asked to read the task description aloud. As described in Section 3.2, participants were asked to imagine being a journalism student with an assignment to write a 30-page term article on a specific topic of debate in U.S. politics. Participants were instructed that the objective of this hypothetical assignment was to give them experience conducting journalistic research using an archival news collection. The task description also specified that they would be given 40 minutes (max) to gather information using a custom-built search system and 15 minutes (max) to create an outline of the hypothetical 30-page term article (i.e., the main deliverable of the task) in an external electronic document.¹ After reading the task description, participants watched a 5-minute video introducing the search system and the tool associated with the task (either the OrgBox or Bookmark tool). Then participants completed a pre-task questionnaire, followed by the search phase and outline generation phase of the task. During the search phase, participants were *not* allowed to use external materials (e.g., pen and paper) to take notes. In other words, they only had access to the search system and provided tools

¹During each task, participants were asked to gather information and generate an outline for a hypothetical 30-page research article. One might ask: Is 55 minutes a realistic timeframe to meet such an objective? During pilot testing, we found 55 minutes to be sufficient. Participants were not expected to plan every detail about the 30-page article. Instead, they were simply expected to create an outline. In doing so, participants had to determine which topics they wanted to cover and how topics and subtopics should be structured. As shown in Figure 5, before the task, participants reported relatively high levels of interest and only moderate levels of expected difficulty.

(OrgBox or Bookmark). After the outline generation phase, participants completed a post-task questionnaire.

3.2 Tasks

Participants completed two tasks that asked them to gather information using a custom-built search system and produce a detailed outline of a 30-page term article. To contextualize our tasks, we provided the following task scenario and objectives for both tasks:

Task scenario: Imagine that you are a student at the UNC School of Media and Journalism. For one of your courses, you have been asked to write a 30-page term article about a specific topic of debate in the United States. The objective of this semester-long project is to give you experience in conducting journalistic research using an archival news collection. To this end, you will be given an access to a collection of news articles published in The Washington Post between 2012–2017.

For your term article, your instructor has asked you to be mindful of the following criteria:

- First, your article should focus on news reporting relevant to the time period covered in the given Washington Post news archive (2012–2017).
- Second, your article should reflect a balance between breadth and depth. In other words, your article should aim at covering different important subtopics in detail.
- Finally, you are encouraged to be creative and write about subtopics that you think would be interesting and important to readers. Such subtopics within a debate may include (but are not limited to): the history of the debate, public opinions, legal issues, social and psychological implications, and so on.

Task Objectives: During the task, you will be asked to search for and organize information on a specific debate. The task will proceed in two stages.

- During the first stage, you will have 40 minutes to search for and gather information using a custom-built system.
- During the second stage, you will have 15 minutes to use the information gathered to produce an outline for your term article. The outline should provide the overall structure and flow of your article. In other words, it should describe which topics (or subtopics) you will write about and in what order. Please also provide a brief description (1–2 sentences) of what information each section (or subsection) will contain.

Participants completed one task about abortion and one task about gun control. The task descriptions are provided below.

Abortion Task: Abortion is probably one of the most controversial subjects in modern society. It involves a number of complex questions including philosophical, legal, and religious issues (among others) related to the deliberate ending of pregnancy before normal childbirth. Public opinions are polarized; they have strong feelings for or against this subject. That's why writing a good article on abortion requires in-depth research of existing newspaper reports. You will have to find a decent number of relevant articles touching on different aspects of the debate in order to create a comprehensive and well-balanced article.

Gun control Task: Gun control (or gun rights) has been considered one of the most controversial debates of our time. It involves a number of complex questions including ideological, legal, and social issues (among others) related to regulating the right of individuals to possess guns. Public opinions are polarized. People have strong feelings for or against this subject. That's why writing a good article on gun control requires in-depth research of existing newspaper reports. You will

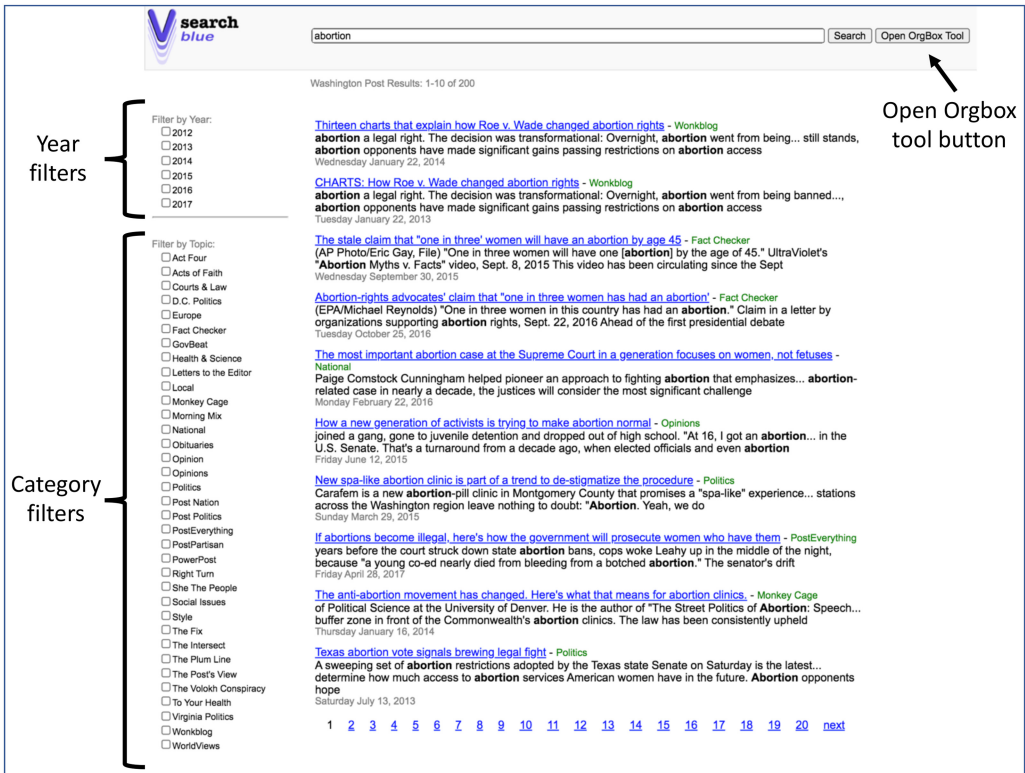


Fig. 2. Search interface.

need to find a decent number of relevant articles touching on different aspects of the debate in order to create a comprehensive and well-balanced article.

3.3 Search Tools

During the study, participants interacted with three tools to search for information and organize the information found. During both experimental tasks, participants interacted with the same search system. To organize the information found, participants used the OrgBox tool during one task and the Bookmark tool during the other.

3.3.1 Search System. To search for information, participants used a custom-built search system (Figure 2). The system was developed using Lucene and provided access to the TREC Washington Post Corpus, which contains 671,947 news articles and blog posts published between January 2012 and December 2017. To retrieve documents, we used the query-likelihood model with Dirichlet smoothing and the default parameter values. The system allowed participants to issue keyword queries and filter search results by publication year and topical category. The search interface also included a button next to the search bar labeled with the name of the tool associated with the condition (i.e., OrgBox or Bookmark). Clicking this button displayed the tool in a pop-up browser window. We used Javascript to log all user interactions (e.g., clicks, scrolls, and mouse-enter/exit events on areas of interest) on all SERPs and clicked landing pages. To capture landing page events, landing pages were dynamically rendered by our system rather than re-directing participants to the online news article hosted by the Washington Post.

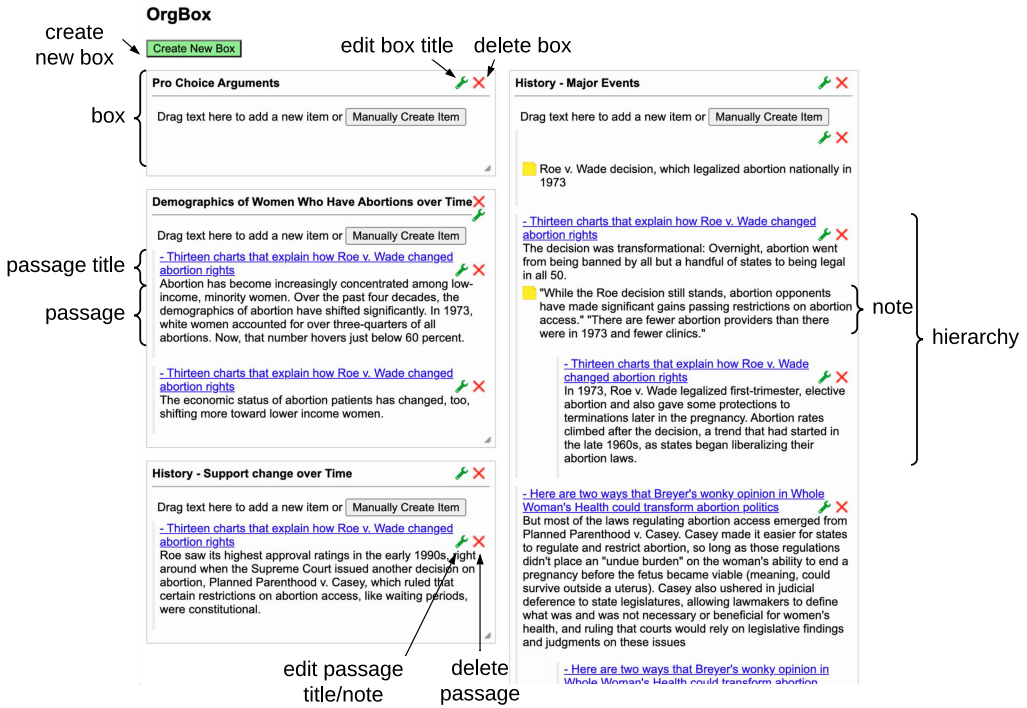


Fig. 3. OrgBox tool.

3.3.2 OrgBox Tool. Figure 3 illustrates the OrgBox knowledge representation tool. The OrgBox tool was designed to help participants externalize their understanding of the assigned topic as they gathered information. Using the OrgBox, participants could not only save the information found but also organize it into clusters (referred to as “boxes”). To create a new box, participants could either click a “create new box” button or drag-and-drop text from a landing page into the OrgBox. Either action triggered a pop-up window that prompted participants to provide a title for the new box. Participants could add “items” to an existing box by drag-and-dropping highlighted text from a landing page into a box. Each added item consisted of three components: (1) the title of the landing page, (2) the textual passage that was drag-and-dropped, and (3) a note added by the participant. When adding a new item to a box, participants were prompted to add a note in a pop-up window. Participants were able to create/add as many boxes and items as desired and were also able to delete and edit boxes and items. Participants could also move items between boxes and organize items within a box in a two-level hierarchy. Finally, participants could move boxes around within the OrgBox. For example, participants could position related boxes closer together or position important boxes in a more salient location. We have released the code for the OrgBox tool as open-source software on our project site² and on Github.³

3.3.3 Bookmark Tool. Figure 4 illustrates the Bookmark tool. The Bookmark tool was designed to serve as a baseline and was, therefore, more rigid than the OrgBox tool. Like the OrgBox tool, participants were able to save information “items” by drag-and-dropping text from a landing

²<http://topaz.ils.unc.edu/demos/orgbox/orgbox.html>.

³<https://github.com/uncinteractivesystemslab/OrgBox>.

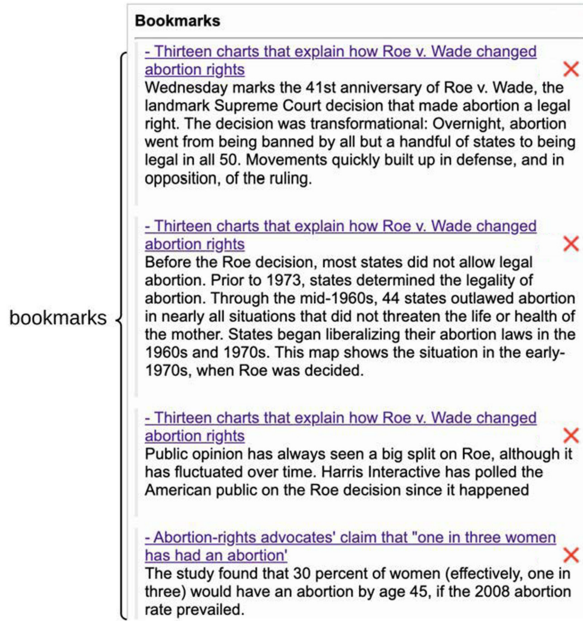


Fig. 4. Bookmark tool (truncated).

page into the Bookmark tool. Each item consisted of two components: (1) the title of the landing page and (2) the text that was drag-and-dropped. Unlike the OrgBox tool, however, participants were not able to organize items into clusters (i.e., boxes). Saved items were stacked vertically in the order in which they were added. Participants were able to delete previously added items, but were not able to make notes about items nor re-order them.

3.4 Questionnaires

Participants completed a pre- and post-task questionnaire before and after each task (i.e., search phase + outline generation phase). In both questionnaires, participants were asked to indicate their level of agreement with statements using a 7-point scale (strongly disagree = 1, strongly agree = 7). The full text of our pre- and post-task questionnaires is available in Appendices A.1 and A.2, respectively.

3.4.1 Pre-task Questionnaire. The pre-task questionnaire included 12 items that measured participants' interest in the task (1 item), prior knowledge (1 item), expected difficulty (4 items), and *a priori* determinability (6 items). The construct of *a priori* determinability relates to the extent to which a searcher knows how to approach the task *before* working on it [11]. Based on Cronbach's alpha, the 6 items for *a priori* determinability had high internal consistency ($\alpha = 0.90$). Thus, we average responses for these items to form one *a priori* determinability measure. The 4 items for difficulty also had high internal consistency ($\alpha = 0.87$). However, we analyzed responses for these items individually so that they could be contrasted with the post-task difficulty measures, which were analyzed individually.

3.4.2 Post-task Questionnaire. The post-task questionnaire was divided into three parts.

The first part included 14 items that measured participants' interest increase in the task domain (1 item), knowledge increase (1 item), experienced difficulty (4 items), satisfaction (4 items), and

self-assessed expertise in the task domain (4 items). Based on Cronbach's alpha, the 4 items for domain expertise had high internal consistency ($\alpha = 0.92$). Thus, we average responses to these items to form one expertise measure. The items for difficulty and satisfaction had slightly lower internal consistency (difficulty = 0.81, satisfaction = 0.82). We treated responses to these items individually because the items revealed interesting differences about which specific aspects of the task were difficult (e.g., the task itself was difficult versus it was difficult to decide when enough information had been found to complete the task.)

The second part of the post-task questionnaire included 10 items that measured participants' perceptions about the quality of the outline produced during the outline generation phase. These items focused on participants' perceptions about the outline having a clearly defined scope, breadth, depth, a balance of breadth and depth, a logical organization, and the extent to which the outline included information supported by articles found during the search phase of the task. Based on Cronbach's alpha, these 10 items had higher internal consistency ($\alpha = 0.90$). Thus, we average responses to these items to form one "outline quality" measure.

Finally, the third part of the post-task questionnaire included 26 items that measured participants' perceptions about the OrgBox or Bookmark tool (depending on the condition) helping them with specific cognitive and metacognitive activities. These items were adapted from those used in McCardle and Hedwin [53]. We included items that measured the extent to which participants perceived the tool to help them with: (1) initial *planning* on how to approach the task (4 items), (2) *connecting* ideas (4 items), (3) *organizing* the information found (4 items), (4) search-oriented *goal-setting* (3 items), (5) *monitoring* their progress (3 items), (6) *evaluating* the information found and their approach to the task (3 items), and (7) *updating* or *revising* their understanding of the task and their goals (5 items). All groups of items had high internal consistency within the group ($\alpha \geq 0.90$). Thus, we averaged responses to these items to form 7 distinct measures.

3.5 Psychometric Tests

During the second session of the study (day two), participants completed two psychometric tests that measured their WM capacity and SW ability. Half the participants' completed the WM test first and the other half completed the SW ability test first. We used computerized tests available with the Inquisit software suite.⁴

3.5.1 Working Memory Test. To measure participants' WM capacity, we used the **Operation Span (OSPAN)** test [18]. During the OSPAN test, participants complete a series of trials. During each trial, participants have presented a visual sequence of letters (ranging from 3–7 letters) one at a time. After each sequence, participants are asked to recall the letters in the order displayed from a 3×3 grid of letters. Additionally, before each letter in a sequence, participants are presented with a relatively simple math problem (e.g., $(8 \times 2) - 8 = ?$) and a proposed solution (e.g., 9). Participants must indicate whether the proposed solution is correct or not. The purpose of these simple math problems is to make it more challenging for participants to memorize the sequence of letters displayed. A participant's final score is equal to the sum of lengths for those sequences perfectly recalled. For example, suppose that a participant perfectly recalled all three letters in a sequence of three, all four letters in a sequence of four, and only four letters in a sequence of five. This participant would receive an OSPAN score of $3 + 4 + 0 = 7$ (i.e., the sequence of five was not perfectly recalled). A higher score indicates that the participant was better at recalling longer sequences. The OSPAN test was used in one of our prior studies [13].

⁴<https://www.millisecond.com/products/inquisit6/laboverview.aspx>.

3.5.2 Switching Ability Test. To measure participants' SW ability, we used the Number-Letter test [55]. During the Number-Letter test, a 2×2 empty grid is displayed on the screen. The test proceeds in a sequence of trials. During each trial, a number-letter pair (e.g., 9F) is displayed in one of the four cells in the grid. The number-letter pairs alternate predictably from cell to cell in the empty grid in a clockwise fashion (i.e., top-left, top-right, bottom-right, bottom-left, top-left). If the number-letter pair appears in the top cells, participants must complete the so-called "letter task"—press the "E" key if the letter is a consonant and "I" if it is a vowel. Alternatively, if the number-letter pair appears in the bottom cells, participants must complete the so-called "number task"—press the "E" key if the number is even and "I" if it is odd. For trials positioned in the top-left or bottom-right cells, participants must *switch* between tasks. In this respect, trials associated with the top-left and bottom-right cells are called *switched* trials (i.e., the current task is different from the previous). Conversely, trials associated with the top-right and bottom-left cells are called *unswitched* trials. A participant's score is equal to the difference between the proportion of correct switched trials minus the proportion of correct unswitched trials. A higher score indicates that the participant is better at switching between number and letter tasks based on the position of the number-letter pair. To our knowledge, the number-letter test has not been used in prior IIR research.

3.6 Search Behaviors

During each task, participants used a custom-built search system to find information and a tool (i.e., the OrgBox or Bookmark) to organize the information found. To investigate RQ1, we logged all user interactions on SERPs, landing pages (rendered by our system), and both tools. We computed the following 23 behavioral measures.

- (1) **Queries:** # of queries.
- (2) **KeywordQueries:** # of distinct keyword queries (explained below).
- (3) **Themes:** # of distinct themes across queries issued (explained below).
- (4) **Clicks:** # of search results clicked.
- (5) **Saves:** # of passages dragged-and-dropped onto the OrgBox or Bookmark tool.
- (6) **QueriesWOClicks:** # of queries without any clicks.
- (7) **QueriesWOSaves:** # of queries without any saved passages.
- (8) **QueriesWOMouseovers:** # of queries without any mouseover events on a search result.
- (9) **QueriesWOScrolls:** # of queries without any SERP scroll events.
- (10) **QueriesWOEvents:** # of queries without any mouseovers, scrolls, nor clicks.
- (11) **QuickReformulations:** # of queries issued within 20 seconds of the previous one.
- (12) **ClicksWOSaves:** # of clicks that did not yield a saved passage.
- (13) **TotalMouseovers:** # of mouseover events on search results.
- (14) **QueriesWithPagnations:** # of queries where the participant examined beyond the top-10 results.
- (15) **SERPScrolls:** total scroll distance on SERPs, measured in SERP-heights.
- (16) **AvgClickRank:** avg. click rank.
- (17) **LandingPageScrolls:** total scroll distance on landing pages, measured in landing page heights.
- (18) **AvgTimeBWEvents:** avg. time difference (in secs.) between SERP events (i.e, queries, clicks, saves).
- (19) **CompletionTime:** session length (in minutes).
- (20) **UniqueQueries:** # of queries not issued by other participants.
- (21) **UniqueKeywordQueries:** # of keyword queries not issued by other participants.
- (22) **UniqueQueryTerms:** # of query terms not used by other participants.
- (23) **UniqueClicks:** # of landing pages not clicked by other participants.

A few of our behavioral measures require explanation. First, in addition to issuing keyword queries, participants could filter search results by year and category using checkboxes displayed on the left (Figure 2). When counting the number of queries (measure #1), we considered each

newly (de-)selected year/category as a new query. Conversely, when computing the number of distinct keyword queries (measure #2), we ignored any years/categories (de-)selected to (re-)filter the search results.

Second, to further investigate participants' search behaviors, we performed a qualitative analysis of all queries issued. For each task (i.e., abortion and gun control), we developed a coding scheme of *themes* represented in all queries. To illustrate, queries issued during the abortion task had themes such as GENERAL (e.g., abortion), COURT CASE (e.g., landmark abortion court cases), ORGANIZATION (e.g., planned parenthood), LEGAL (e.g., state restrictions on abortion), STATISTICS (e.g., abortion rates), and HEALTH (e.g., pregnancy timeline). Queries issued during the gun control task had themes such as GENERAL (e.g., gun control), INTERNATIONAL (e.g., gun control outside of America), POLITICS (e.g., politics and gun reform), VIOLENCE (e.g., sandy hook shooting), ORGANIZATION (e.g., national rifle association), and CULTURE (e.g., white supremacy and guns). To validate our coding scheme of themes, two of the authors independently coded all queries from five participants (about 16% of the data). Using Cohen's Kappa, the inter-coder agreement was $\kappa = 0.873$, which is considered "almost perfect" agreement [47]. Given, this high agreement level, one of the authors coded the remaining 87% of the data. Using this qualitative coding of queries, we computed measure #3—the number of distinct themes across queries in the search session.

Finally, measures #20–23 capture the extent to which a participant adopted search strategies that were *different* from other participants for the same task. Measure #20 corresponds to the number of queries that were not issued by other participants, considering the keywords used *and* any years/categories selected to filter the search results. Conversely, measure #21 considers only the keyword portion of the queries issued.

Our 23 behavioral measures were designed to capture different types of behaviors. Measure #1 captured the number of queries issued; measures #2–3 captured the level of query-based exploration; measures #4–5 captured clicks and saves; measures #6–11 captured query abandonment; measure #12 captured click abandonment; measures #13–16 captured the level of SERP exploration; measure #17 captured the level of landing page exploration; measure #18 captured the pace of interaction; measure #19 captured the task completion time; and measures #20–23 captured the extent to which the participant adopted search strategies that were different from other participants.

3.7 Outline Assessment

During the *outline generation phase* of the study, participants were asked to produce an outline of their hypothetical 30-page assignment using the information gathered with the OrgBox or Bookmark tool. Participants produced their outlines using a Google Doc. To help standardize the format of outlines produced, the Google Doc contained a hierarchical bulleted list with placeholders as a starting point (e.g., main topic 1, subtopic 1.1, sub-subtopic 1.1.1).

In RQ3, we investigate the effects of the interface condition, WM capacity, and SW ability on the quality of outlines produced by participants. To this end, we developed a qualitative coding scheme to evaluate outlines across six dimensions.

- (1) BREADTH: the extent to which an outline covers a broad range of topics.
- (2) DEPTH: the extent to which an outline explores at least one topic in great detail.
- (3) BALANCE: the extent to which an outline covers all topics with a similar level of detail. For example, an outline would score low on this dimension if it provides detailed coverage of some topics but not others.
- (4) ORG1: the extent to which the main topics of the outline (usually the highest level bullet points) have logical transitions from one to the next and the same level of abstraction.

Table 1. Descriptive Statistics of Participants' Scores on Both Psychometric Tests

	Working Memory	Switching
Min	0	-0.25
Max	75	0.05
Mean	39.97	-0.07
SD	19.61	0.08
Median	38.50	-0.05
Participants in Low Group	16	17
Participants in High Group	16	15

- (5) ORG2: the extent to which the subtopics (and sub-subtopics) within each main topic have logical transitions from one to the next and similar levels of abstraction.
- (6) ORG3: the extent to which an outline contains cohesive sets of subtopics. For example, an outline would score low on this dimension if it contains many *loosely* tied subtopics (level 2–3 bullet points) within the same main topic.

Outlines were rated by one of the authors along these six dimensions using two categories: low vs. high. To test the reliability of our coding scheme, two of the authors rated outlines from six participants (about 19% of all outlines). The Cohen's κ agreement was: BREADTH = 0.600, DEPTH = 0.455, BALANCE = 0.478, ORG1 = 0.765, ORG2 = 0.454, and ORG3 = 0.640. Agreement ranged from 'moderate' to 'substantial' [47]. As reported in Section 4.4, our results found only one significant effect on the quality of outlines produced by participants. Using the OrgBox, participants produced outlines with more coherent sets of subtopics (Org3). For the Org3 dimension, inter-coder agreement was 'substantial' (i.e., $\kappa > 0.60$) [47].

3.8 Data Analysis

In this section, we report on participants' WM and SW ability scores. Additionally, we explain two decisions made in our analysis: binning participants into low- and high-WM/SW groups and using multilevel modeling for our statistical analyses.

3.8.1 Psychometric Test Scores. Table 1 shows descriptive statistics of participants' scores on both psychometric tests. To analyze the effects of each cognitive ability, we decided to group participants into low and high groups using a median split. Table 1 shows the number of participants assigned to each group for WM and SW ability. Prior IR studies on the effects of cognitive abilities have also binned participants using either a median split [2, 3, 9, 13, 74] or by comparing between participants in the bottom and top quartiles [32]. The Pearson's correlation between participants' scores on both tests was $r = 0.143$ ($p = .436$). A low correlation suggests that both psychometric tests are measuring different cognitive abilities. After binning participants into low/high groups, 19 participants (60%) were assigned to either both low groups (i.e., low-WM and low-SW) or both high groups (i.e., high-WM and high-SW). The other 13 participants (40%) were assigned to the low group for one ability and the high group for the other.

3.8.2 Statistical Analysis. To address RQ1–RQ3, we were interested in measuring the following main and interaction effects: (1) the main effect of the interface condition (1 main effect), (2) the main effect of each cognitive ability (2 main effects), and (3) the two-way interaction effects between the interface condition and each cognitive ability (2 interaction effects). In our statistical analysis, we decided to use multilevel modeling.

Multilevel models (MLMs) are similar to other regression techniques. However, MLMs are particularly powerful when the data are “grouped” and we wish to account for *random* effects introduced by these groups. In our case, each participant completed two tasks and so our data was grouped by the participant. In our models, we wanted to account for random effects at the participant level. For example, random effects might be due to individual differences beyond WM and SW ability (e.g., prior knowledge, search experience). To model random effects at the participant level, we used the participant ID as a *random intercept* factor in all our models.

Each of our models included six predictors (i.e., fixed factors): one for task topic, one for interface, one for WM capacity group (low- vs. high-WM), one for SW ability group (low- vs. high-SW), and one for each two-way interaction between the interface and each cognitive ability group. As described in Section 3.2, participants completed one task on abortion and one on gun control. While we aimed at designing tasks with similar characteristics, participants reported lower prior knowledge for the gun control task (details in Section 4.1). Thus, we decided to include task ID as a covariate. All our independent variables were categorical (vs. real-valued). Thus, independent variables were included in our MLMs as indicator variables. For task ID, abortion was coded as “0” and gun control was coded as “1”. For interface, the Bookmark tool was coded as “0” and the OrgBox tool was coded as “1”. Finally, for each cognitive ability, the low group was coded as “0” and the high group as “1”. In this respect, positive β values indicate a positive effect on the corresponding outcome measure from the gun control task, OrgBox tool, and high cognitive ability group. To further explain, each of our models (one per dependent variable) used the following equation:

$$Y_{ij} = \underbrace{\beta_0 + \mu_j}_{\text{random intercept}} + \underbrace{\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_6 X_6}_{\text{fixed factors}} + \underbrace{e_{ij}}_{\text{random error}},$$

where Y_{ij} denotes the outcome variable for the i th data point and j th participant. As noted, the y -intercept in our models was a linear combination of parameter β_0 (a global parameter) and μ_j (specific to participant j). Parameters β_1 – β_6 (shared amongst all participants or “groups”) denote the β values for our six fixed factors.

4 RESULTS

4.1 Preliminaries: Effects of Task on Pre-task Perceptions

Before reporting on results for RQ1–RQ3, we report on the effects of the task topic (abortion vs. gun control) on participants’ pre-task perceptions. In the pre-task questionnaire, participants were asked about their level of interest in the task topic, prior knowledge, *a priori* determinability, and expected difficulty. In terms of difficulty, they were asked about difficulty in searching for information, integrating the information found, deciding when they had enough information, and overall difficulty. Figure 5 shows the differences in participants’ perceptions between both task topics. While we aimed at designing tasks with similar characteristics, our gun control task was expected to be slightly more challenging than our abortion task. Even before performing the task, participants reported lower interest and slightly greater expected difficulty for the gun control topic. Additionally, task topic had a significant effect on prior knowledge ($\beta = -0.700$, S.E. = 0.257, $p = .010$). Given this effect of tasks, we decided to include tasks as a covariate in our MLM analyses for RQ1–RQ3.

4.2 RQ1: Effects on Search Behaviors

In Sections 4.2–4.4, we present our results for RQ1–RQ3, respectively. In each section, we first report on the main effects of the interface condition (i.e., OrgBox vs. Bookmark). Then we report on the effects of WM and SW, both their main effects and two-way interaction effects with interface

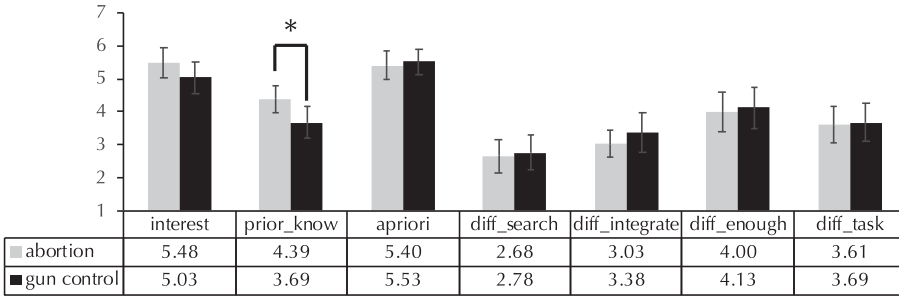


Fig. 5. The effects of task on participants' pre-task perceptions (Means and 95% CIs). "*" denotes a significant difference at the $p < .05$ level.

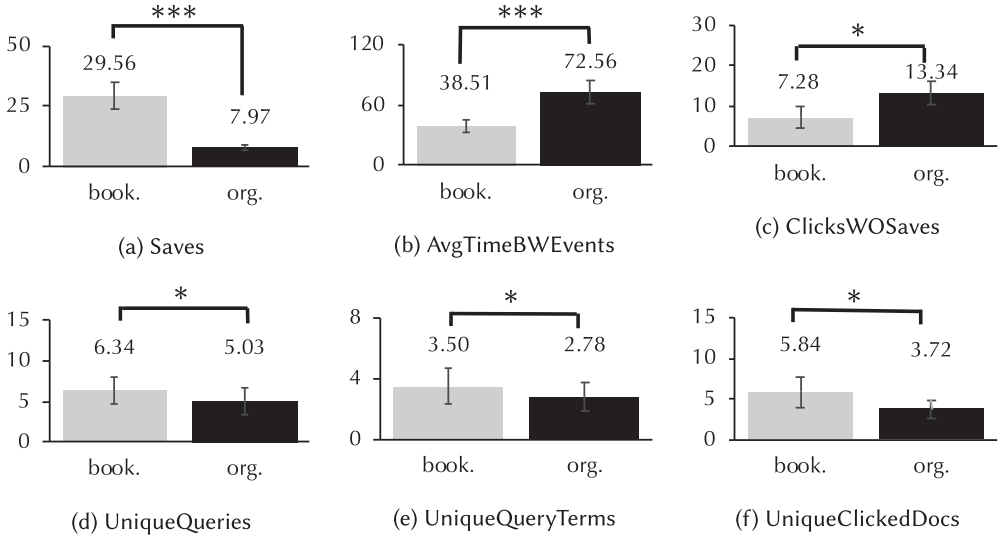


Fig. 6. Effects of interface on search behaviors (Means and 95% CIs). "*" and "***" denote significant differences at the $p < .05$ and $p < .001$ level.

(i.e., $WM \times \text{interface}$ and $SW \times \text{interface}$). Following each set of results, we provide potential interpretations of the observed trends. We elaborate on these interpretations in Section 5. In RQ1, we investigate the effects of the interface (OrgBox vs. Bookmark), WM capacity, and SW ability on participants' search behaviors.

4.2.1 Main Effects of Interface Condition. The interface had a significant main effect on six behavioral measures. As shown in Figure 6, when using the OrgBox (vs. Bookmark), participants: (1) saved fewer passages ($\beta = -21.836$, S.E. = 3.784, $p = .000$); (2) took longer time between search interactions ($\beta = 39.777$, S.E. = 7.920, $p = .000$); (3) had more clicks without any saves ($\beta = 5.523$, S.E. = 2.082, $p = .012$); (4) issued fewer queries not issued by other participants ($\beta = -2.735$, S.E. = 1.063, $p = .015$); (5) used fewer query terms not used by other participants ($\beta = -1.282$, S.E. = 0.628, $p = .049$); and (6) clicked fewer results not clicked by other participants ($\beta = -2.546$, S.E. = 1.215, $p = .044$).

Interpretation: We observed three main trends. First, using the OrgBox, participants searched at a slower pace. One possibility is that the OrgBox required participants to perform more tasks

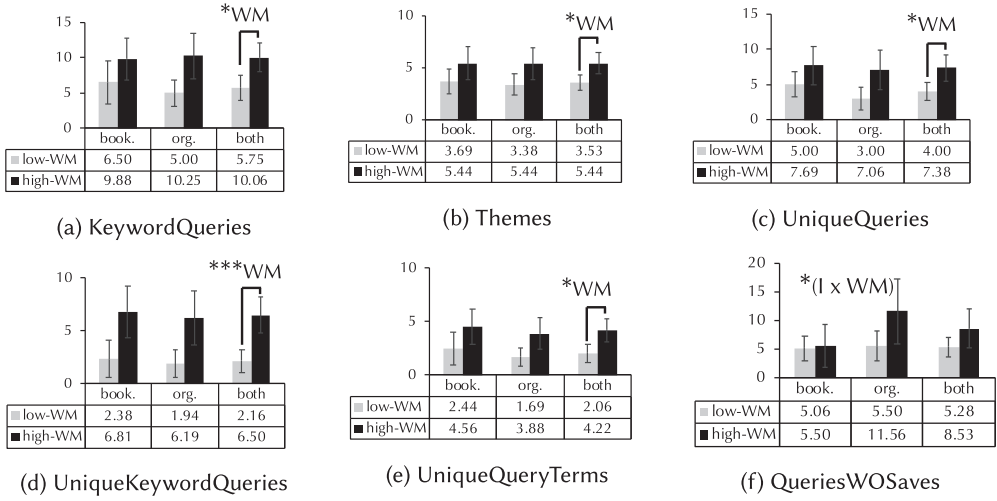


Fig. 7. Effects of WM on search behaviors (Means and 95% CIs). “*” and “***” denote significant differences at the $p < .05$ and $p < .001$ level.

outside of searching (e.g., create boxes, organize boxes, make notes). Second, participants saved fewer items and were more deliberate (e.g., had more clicks without saves) with the OrgBox. These trends suggest that the OrgBox influenced participants search for specific information to add to the current set of boxes. Finally, with the OrgBox, participants issued more queries with similar characteristics as those issued by other participants. One possibility is that participants issued more general queries and explored different subtopics (i.e., boxes) in parallel. To elaborate, the Bookmark tool did not allow participants to organize passages by topic. Therefore, with the Bookmark tool, participants were more likely to explore subtopics *serially* as a “workaround”, influencing them to issue more specific queries that were different from those issued by other participants.

4.2.2 Working Memory: Main and Interaction Effects. WM had five main effects and one interaction effect on behavioral measures. As shown in Figure 7(a)–(e), regardless of the interface, high-WM participants: (1) issued more distinct keyword queries ($\beta = 4.357$, S.E. = 1.901, $p = .026$); (2) issued queries with a greater variety of themes ($\beta = 2.182$, S.E. = 0.898, $p = .019$); (3) issued more queries not issued by other participants ($\beta = 3.197$, S.E. = 1.577, $p = .048$); (4) issued more keyword queries not issued by other participants ($\beta = 5.236$, S.E. = 1.375, $p = .000$); and (5) used more query terms not used by other participants ($\beta = 2.519$, S.E. = 0.944, $p = .011$).

Additionally, WM and interface had a significant interaction effect on the number of queries without saves ($\beta = 5.263$, S.E. = 2.476, $p = .041$). As shown in Figure 7(f), with the Bookmark tool, low- and high-WM participants had a comparable number of queries without saves. Conversely, with the OrgBox tool, high-WM participants had more queries without saves than low-WM participants.

Interpretation: We observed two main trends. First, high-WM participants issued queries targeting a broader range of subtopics (e.g., issued more keyword queries, more queries not issued by others, and more queries exploring a wider variety of themes). Second, with the OrgBox, high-WM participants had more queries without any saves. This result suggest that, with the OrgBox, high-WM participants were more deliberate and selective in choosing passages to save than low-WM participants.

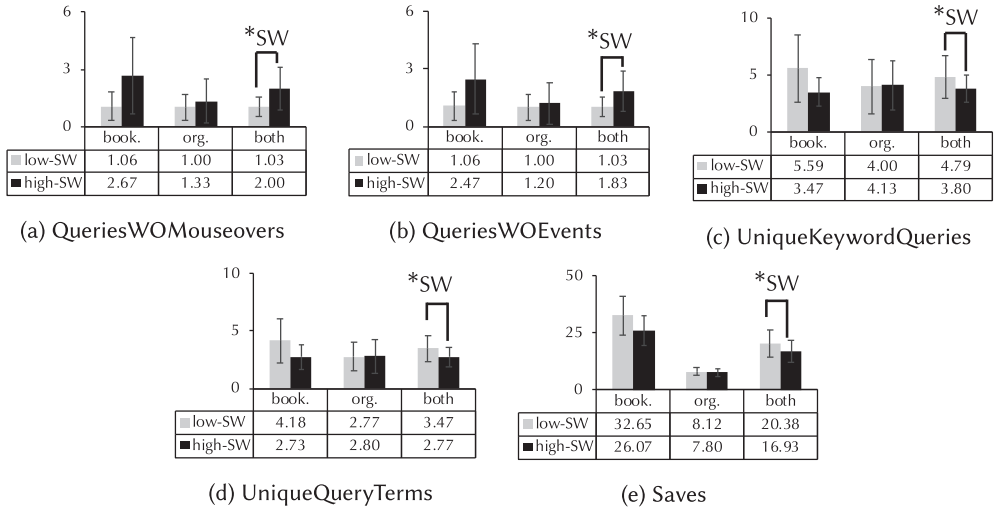


Fig. 8. Effects of SW ability on search behaviors (Means and 95% CIs). “*” denotes a significant difference at the $p < .05$ level.

4.2.3 Switching Ability: Main and Interaction Effects. SW ability had a significant main effect on five behavioral measures. As shown in Figure 8, regardless of the interface, high-SW participants: (1) issued more queries without any mouseover events ($\beta = 1.750$, S.E. = 0.802, $p = .033$); (2) issued more queries without any events ($\beta = 1.531$, S.E. = 0.754, $p = .047$); (3) issued fewer keyword queries not issued by other participants ($\beta = -3.050$, S.E. = 1.367, $p = .030$); (4) used fewer query terms not used by other participants ($\beta = -1.908$, S.E. = 0.941, $p = .048$); and (5) had fewer saves ($\beta = -7.842$, S.E. = 3.690, $p = .038$).

Interpretation: We observed two main trends. First, high-SW participants had more abandoned queries and fewer saves. This trend suggests that high-SW participants had stricter relevance criteria, perhaps because they leveraged the tool more to decide what to search for and save. Second, high-SW participants issued fewer unique keywords queries and used fewer unique query terms. One possibility is that their higher SW ability influenced high-SW participants to issue broader queries and explore different subtopics in parallel.

4.3 RQ2: Effects on Post-task Perceptions

In RQ2, we investigate the effects of the interface (OrgBox vs. Bookmark), WM capacity, and SW ability on participants’ post-task perceptions.

4.3.1 Main Effects of Interface Condition. Interface had a significant main effect on four post-task measures. As shown in Figure 9, when using the OrgBox (vs. Bookmark), participants reported greater difficulty ($\beta = 1.194$, S.E. = 0.454, $p = .013$), as well as a greater sense that the tool helped them with (meta)cognitive activities related to planning ($\beta = 1.114$, S.E. = 0.417, $p = .012$); organizing ($\beta = 1.947$, S.E. = 0.522, $p = .001$); and monitoring ($\beta = 1.187$, S.E. = 0.404, $p = .006$).

Interpretation: By allowing participants to form subtopic clusters (i.e., boxes) and by having this visual representation accessible, the OrgBox was perceived to help participants plan their searches, organize the information found, and monitor their progress. There are several possibilities for why the OrgBox also made the task more difficult. First, compared to the Bookmark tool, the OrgBox had more functions to learn. Second, the OrgBox may have influenced

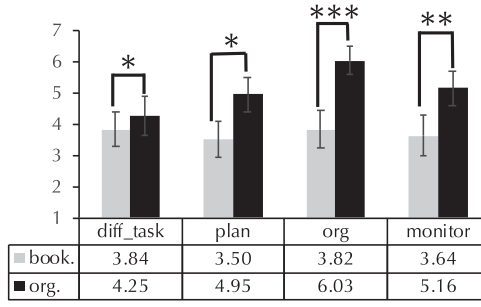


Fig. 9. The effects of the interface on participants' post-task perceptions (Means and 95% CIs). “*”, “**”, and “***” denote significant differences at the $p < .05$, $p < .01$, and $p < .001$ level.

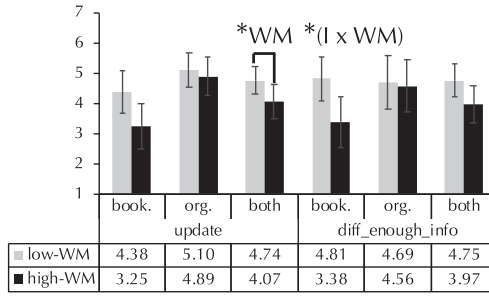


Fig. 10. The effects of WM on participants' post-task perceptions (Means and 95% CIs). “*”, “**”, and “***” denote significant differences at the $p < .05$, $p < .01$, and $p < .001$ level.

participants to perform more targeted searches—searching with more specific intentions (e.g., for specific boxes). Third, having the boxes visually accessible may have raised participants' awareness of gaps in knowledge (i.e., monitoring). Finally, the OrgBox may have influenced participants to address subtopics in parallel versus in series. With the OrgBox, participants could add passages to different boxes as they searched, which requires considering multiple subtopics at a time.

4.3.2 Working Memory: Main and Interaction Effects. WM had one main effect and one interaction effect on post-task perceptions. As shown in Figure 10, WM had a main effect on participants' perceptions of the tool helping them with (meta)cognitive activities related to updating ($\beta = -1.083$, S.E. = 0.463, $p = .022$), with low-WM participants reporting higher levels than high-WM participants. Additionally, WM and interface had a significant interaction effect on participants' perceptions of difficulty in determining when they had *enough information* to complete the task ($\beta = 1.524$, S.E. = 0.672, $p = .030$). As shown in Figure 10, low-WM participants reported greater difficulty than high-WM participants when using the Bookmark to gather information. Conversely, both participant groups reported similar levels of difficulty when using the OrgBox.

Interpretation: Compared to high-WM participants, low-WM participants perceived *both* tools to help with metacognitive updating, which included revising their approach to the task and/or their understanding of the task domain. In other words, low-WM participants perceived even the Bookmark tool to be helpful for updating. Additionally, with the OrgBox, low- and high-WM participants reported similar levels of difficulty in determining when they had enough information to complete the task. Conversely, with the Bookmark, low-WM participants reported greater levels of difficulty than high-WM participants. The OrgBox allowed low- and high-WM participants to

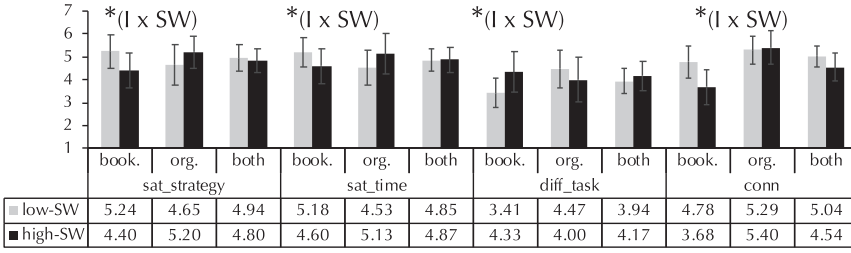


Fig. 11. The effects of SW ability on participants' post-task perceptions (Means and 95% CIs). “*” denotes a significant difference at the $p < .05$ level.

easily monitor the subtopics covered (i.e., boxes) and the number of saved passages for each subtopic. The Bookmark did not allow participants to visually organize passages by subtopic. Thus, participants had to rely more on their WM to monitor the amount of content across subtopics, which may have been more difficult for low-WM participants.

4.3.3 Switching Ability: Main and Interaction Effects. SW and interface had significant interaction effects on four post-task measures: (1) satisfaction with the search strategy ($\beta = 1.431$, S.E. = 0.669, $p = .040$); (2) satisfaction with the time allotted for the task ($\beta = 1.204$, S.E. = 0.486, $p = .019$), (3) the overall difficulty of the task ($\beta = -1.272$, S.E. = 0.568, $p = .032$), and (4) the extent to which the tool helped participants with connecting ideas ($\beta = 1.186$, S.E. = 0.515, $p = .028$).

Figure 11 shows two main trends. First, in terms of satisfaction and perceptions of difficulty, high-SW participants had a better experience with the OrgBox and low-SW participants had a better experience with the Bookmark. With the OrgBox tool, high-SW participants reported greater satisfaction with their search strategy and the allotted time, as well as lower overall difficulty. Conversely, low-SW participants had the opposite trend—they reported greater satisfaction and lower difficulty with the Bookmark tool.

Second, with both tools, low-SW participants reported similar perceptions about the tool helping them connect ideas. Conversely, high-SW participants reported higher perceptions about the tool helping them connect ideas with the OrgBox vs. Bookmark.

Interpretation: With the OrgBox, participants had to switch between different types of cognitive and physical activities, including searching for and judging the relevance of information, organizing the information saved, and monitoring their progress across subtopics, planning next steps, and so on. It is possible that high-SW participants perceived this switching requirement of the OrgBox to be less burdensome and were better able to leverage the OrgBox to connect ideas.

4.4 RQ3: Effects on Outline Evaluation

In RQ3, we investigate the effects of the interface (OrgBox vs. Bookmark), WM capacity, and SW ability on the quality of outlines produced by participants. As described in Section 3.7, outlines were assessed along six dimensions: (1) BREADTH, (2) DEPTH, (3) BALANCE, (4) ORG1, (5) ORG2, and (6) ORG3. Along each dimension, outlines were coded as either “low” or “high” quality. To analyze the effects on the quality of outlines, we used multilevel logistic regression models. Low quality was coded as a “0” and high quality was coded as a “1”.

4.4.1 Main Effects of Interface Condition. As shown in Figure 12, participants produced better outlines using the OrgBox (vs. Bookmark) across all six dimensions. The interface had a significant effect on ORG3 ($\beta = 4.843$, S.E. = 1.864, $p = .009$). Using the OrgBox, participants produced outlines with more cohesive sets of subtopics.

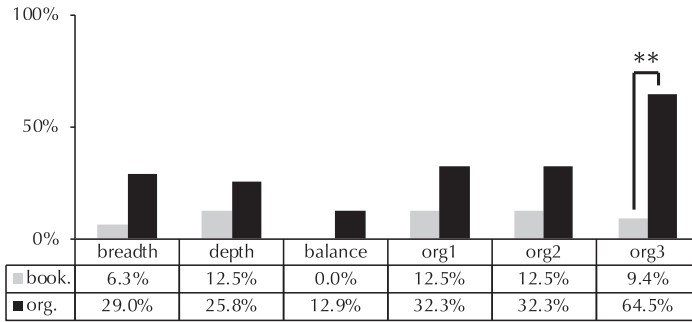


Fig. 12. The effects of the interface on the quality of outlines produced by participants. Percentage of participants whose outlines were judged as “high” quality. “***” denotes a significant difference at the $p < .01$ level.

Interpretation: The OrgBox allowed participants to *explicitly* organize information by subtopics or themes (i.e., different boxes). Our results suggest that the OrgBox helped participants define categories and generate a more organized representation of the task domain. Additionally, this external representation of the domain was available during the outline generation phase. Given these benefits, participants were able to produce outlines with more cohesive sets of subtopics.

WM and SW ability had no significant main effects nor interaction effects with the interface condition. Our interpretation is that low-WM and low-SW participants were able to leverage the OrgBox to produce outlines of similar quality as high-WM and high-SW participants.

4.4.2 Working Memory: Main and Interaction Effects. WM did not have a significant main effect nor interaction effect on any outline quality measure.

4.4.3 Switching Ability: Main and Interaction Effects. SW ability did not have a significant main effect nor interaction effect on any outline quality measure.

5 DISCUSSION

In this section, we summarize our results, compare them with results from prior work, and discuss their implications. Table 2 summarizes the main trends observed in our results.

5.1 Main Effects of Interface

The interface condition had significant effects on participants’ search behaviors (RQ1), post-task perceptions (RQ2), and the quality of outlines (RQ3). In a previous article [12], we reported on a qualitative analysis of an exit interview with 16 (out of 32) participants from this same study. The objective of this exit interview was to gain insights about participants’ perceptions of: (1) the benefits and challenges introduced by each tool (OrgBox vs. Bookmark) and (2) the influences of each tool on the approaches taken by participants to complete the task. Below, in discussing our RQ1–RQ3 results, we also consider how they are consistent with comments made by participants during the exit interviews.

Search Behaviors (RQ1): In terms of search behaviors (Figure 6), participants searched differently depending on the interface. With the OrgBox, participants had more to do besides search. Participants could create/edit/delete boxes, (re-)organize items, and make/edit notes. Thus, as might be expected, participants took longer between search-related events when using the OrgBox. With the OrgBox, participants also had to *classify* saved passages by adding them to an existing box or a

Table 2. Summary of Results

	Search Behaviors (RQ1)	Post-task Perceptions (RQ2)	Outline Quality (RQ3)
Interface (I)	– The OrgBox tool influenced participants to: (1) search at a slower pace, (2) be more selective when adding information to the tool, and (3) explore subtopics in parallel using general queries.	– Participants perceived the OrgBox to help with meta-cognitive “organizing”, “planning”, and “monitoring”. – Using the OrgBox, participants perceived the task to be more difficult.	– Using the OrgBox, participants produced outlines with more cohesive sets of subtopics.
Working Memory (WM)	– High-WM participants issued queries that targeted a wider-range of subtopics (e.g., more keyword queries, themes, and unique queries/query-terms).	– Low-WM participants perceived both tools to help with meta-cognitive “updating”.	N/A
Switching Ability (SW)	– High-SW participants were more selective (e.g., more abandoned queries, fewer saves) and issued fewer queries with unique characteristics.	N/A	N/A
I x WM	– The OrgBox influenced participants to be more selective and this effect was <i>stronger</i> for high-WM participants (e.g., more queries without saves).	– With the Bookmark tool, low-WM participants reported greater difficulty in determining when they had enough information to complete the task.	N/A
I x SW	N/A	– Using the OrgBox (vs. Bookmark), high-SW participants reported greater satisfaction (search strategy and allotted time) and lower difficulty. This trend was the opposite for low-SW participants. – Using the OrgBox (vs. Bookmark), high-SW participants perceived the tool to provide greater support for meta-cognitive “connecting”. This trend was not observed for low-SW participants.	N/A

new box. This influenced participants to be more deliberate and selective when saving passages to the tool. Finally, using the OrgBox, participants exhibited less divergent search strategies (i.e., had fewer unique queries, query terms, clicked results). This could be because the OrgBox tool allowed participants to explore subtopics in parallel by issuing general queries (e.g., abortion, gun policy). In contrast, with the Bookmark, participants could neither group nor re-order the saved passages by subtopic. This feature of the Bookmark tool influenced participants to explore subtopics *serially* versus in parallel, which might have led to more divergent querying and clicking behavior. Indeed, Choi et al. [12] found that the inability to organize information in the Bookmark tool influenced some participants to explore subtopics serially as a “workaround”.

Post-task Perceptions (RQ2): In terms of post-task perceptions (Figure 9), participants perceived that the OrgBox helped them with (meta)cognitive activities related to organizing, planning, and monitoring. The effect on organizing is perhaps not surprising—the OrgBox tool was designed to help searchers organize saved passages. It is noteworthy, however, that participants perceived the OrgBox to help them with planning their searches and monitoring their progress towards specific goals during the search. In Choi et al. [12], several participants commented that the OrgBox influenced them to plan their searches and monitor the breadth and depth of information gathered. This result is also consistent with research in the learning sciences, which has found that external knowledge representations can encourage and support important (meta)cognitive activities such as goal-setting and progress monitoring [78].

Using the OrgBox, participants also perceived the task to be more difficult. Here, we see two possible explanations. First, as stated previously, participants simply had more to do with the OrgBox. Second, by eliciting more planning and monitoring, the OrgBox influenced participants to search more deliberately (i.e., planning), as well as evaluate their progress towards specific goals and maintain awareness of knowledge gaps (i.e., monitoring). The additional physical and cognitive work induced by the OrgBox might have led to increased perceptions of task difficulty. However, we believe that a greater perception of difficulty is not always a negative outcome. Greater difficulty may also reflect more ambitious objectives and greater effort towards these objectives.

Outline Quality (RQ3): In terms of the quality of outlines (Figure 12), while the OrgBox made the task more difficult, it also improved the quality of the task’s main deliverable. Along all six dimensions considered, participants produced better outlines with the OrgBox. Specifically, participants produced outlines with more cohesive sets of subtopics. Here, we see two possible explanations. First, using the OrgBox, participants organized saved passages into subtopics during the search phase. Thus, it is possible that the OrgBox helped participants produce better outlines by influencing them to think about the outline structure earlier in the task (i.e., during the search phase) rather than later (i.e., during the outline generation phase). Second, during the outline generation phase, it is likely that the visual structure of the OrgBox (and added notes) made it easier for participants to build cohesive clusters of subtopics (and sub-subtopics) in their outlines. In Choi et al. [12], several participants mentioned that the visual and textual representations in the OrgBox served as a “draft” of their outline.

5.2 Effects of Working Memory Capacity

WM capacity had significant effects on participants’ search behaviors (RQ1) and post-task perceptions (RQ2). In some cases, the effect of WM was *conditioned* on the interface (i.e., an interaction effect). WM had no significant effects on the quality of outlines (RQ3).

Search Behaviors (RQ1): In terms of search behaviors (Figure 7), we observed two main trends. First, high-WM participants explored a larger space than low-WM participants. Regardless of the interface, high-WM participants issued queries that explored a broader range of themes (Figure 7(a)–(b)) and issued more queries that diverged from those of other participants

(Figure 7(c)–(e)). This result is largely consistent with prior studies, which have found that high-WM searchers tend to exert more effort and satisfice less than low-WM searchers [13, 31, 32]. Second, WM and interface had a significant interaction effect on the number of queries without a saved passage. With the OrgBox, high-WM participants had more queries without a save than low-WM participants. This trend was not observed with the Bookmark tool. As mentioned in Section 5.1, the OrgBox influenced participants to search more deliberately, and this effect was *more pronounced* for high-WM (vs. low-WM) participants. This trend resonates with results from Fourney et al. [27], who compared relevance judgments from participants with versus without dyslexia, a medical diagnosis associated with deficits in WM capacity. In that study, non-dyslexic participants produced relevance judgments that were more bimodal (i.e., either highly relevant or not relevant). Conversely, dyslexic participants produced relevance judgments with a more central tendency. In our case, the OrgBox may have influenced high-WM participants to have more specific (or stricter) relevance criteria than low-WM participants, leading to more queries without a saved passage.

Post-task Perceptions (RQ2): In terms of post-task perceptions (Figure 10), we observed two main trends. First, regardless of the interface, low-WM participants reported higher perceptions about the tool helping them with metacognitive activities related to updating—changing the approach to the task or their understanding of the task topic. It is possible that high-WM searchers were better able to perform such metacognitive activities without the help of the tool. Second, WM and interface had a significant interaction effect on participants’ perceptions of difficulty in determining when they had enough information to complete the task. With the Bookmark, low-WM participants reported greater difficulty than high-WM participants. Compared to the OrgBox, the Bookmark tool provided no visual cue about the topical coverage of saved passages. Thus, it is possible that low-WM participants found it more difficult to keep track of the topical coverage of saved passages in short-term memory.

5.3 Effects of Switching Ability

SW ability had significant effects on participants’ search behaviors (RQ1) and post-task perceptions (RQ2). In some cases, the effect of SW was *conditioned* on the interface (i.e., an interaction effect). SW had no significant effects on the quality of outlines (RQ3).

Search Behaviors (RQ1): In terms of search behaviors (Figure 8), SW had a significant main effect on five measures. Prior research has not investigated the effects of SW ability in IIR, particularly in scenarios where searchers need to multitask between searching and saving/structuring information. Thus, while we provide possible explanations for our results, further investigation is needed to verify whether they represent generalizable trends.

Our results suggest two main trends. First, regardless of the tool, high-SW participants had more abandoned queries (Figure 8(a)–(b)) and fewer saved passages (Figure 8(e)). This suggests that high-SW participants searched more deliberately and had stricter relevance criteria. This may be because high-SW participants were better able to shift their attention between searching, referring to the OrgBox or Bookmark to remind themselves of specific subgoals, and evaluating the information found. Second, high-SW participants issued fewer queries that diverged from those of other participants (Figure 8(c)–(d)). This suggests the possibility that high-SW participants explored subtopics in parallel by issuing broader queries (e.g., abortion, gun policy). In other words, it is possible that their ability to switch between different perspectives more effortlessly and effectively influenced high-SW participants to explore topics in parallel.

Post-task Perceptions (RQ2): In terms of post-task perceptions (Figure 10), SW ability and interface had four significant interaction effects. Our results suggest that high-SW participants had a better experience using the OrgBox and that low-SW participants had a better experience using

the Bookmark. High-SW participants reported greater levels of satisfaction and slightly lower levels of difficulty with the OrgBox (vs. Bookmark). Conversely, low-SW participants reported greater levels of satisfaction and lower levels of difficulty with the Bookmark (vs. OrgBox). Additionally, high-SW participants reported greater perceptions that the OrgBox (vs. Bookmark) helped them with metacognitive activities related to *connecting*—connecting ideas and considering how parts of the topic related to each other. Using the OrgBox, participants had to multitask between different types of physical and cognitive activities—searching for information and externalizing their knowledge structures. Our results suggest that high-SW participants found these multitasking activities less burdensome and were better able to leverage the OrgBox to connect ideas within the task domain.

5.4 Implications

Our results have several important implications.

First and foremost, our results suggest that tools such as the OrgBox can provide important benefits. In terms of RQ1, the OrgBox tool influenced participants to search more deliberately. In terms of RQ2, participants reported that the OrgBox helped them with (meta)cognitive activities related to organizing, planning, and monitoring. It is noteworthy that participants perceived the tool to be helpful for, not only organizing information but also for planning their searches and monitoring their progress. In other words, by allowing participants to *externalize* their knowledge structures, they perceived the OrgBox to also help them define goals and track their progress towards their goals. Finally, in terms of RQ3, using the OrgBox, participants produced higher-quality outlines (the task's main deliverable). Specifically, using the OrgBox, participants produced outlines with more cohesive sets of subtopics. We believe that the OrgBox influenced participants to think about the structure of their outlines earlier in the task (i.e., during the search phase) rather than postponing this activity until the outline generation phase. Together, our results suggest that tools to support complex search tasks should encourage users to externalize their knowledge representations during the search process. Our RQ2 and RQ3 results resonate with research in self-regulated learning, which has found that goal-setting can influence learners to expend more effort and achieve better performance [48, 82].

Secondly, our results suggest that tools such as the OrgBox can provide unique benefit for searchers with specific abilities. As previously mentioned, we believe that the OrgBox tool influenced participants to search more deliberately (i.e., with specific goals and/or relevance criteria in mind). Our results suggest that this trend was slightly stronger for high-WM participants. As shown in Figure 7(f), with both tools, low-WM participants issued a similar number of queries that did not yield a save (i.e., evidence of purposeful searching). Conversely, with the OrgBox, high-WM participants issued more queries that did not yield a save.

Third, our results also suggest that tools such as the OrgBox can introduce unique challenges for searchers with specific abilities. In our study, participants reported different perceptions based on their SW ability. Using the OrgBox, low-SW participants reported lower satisfaction and greater difficulty. Conversely, using the OrgBox, high-SW participants reported greater satisfaction and perceptions of the tool being helpful for connecting ideas. This result emphasizes the claim that “one size does not fit all” [68]. Tools such as the OrgBox, which require more multitasking, may introduce challenges to searchers with lower SW ability. Future research should explore how search assistance tools can be designed to better support such searchers.

Finally, our research adds to the literature on how cognitive abilities influence search behavior. Consistent with prior work, our results show that higher WM searchers exert more effort and search more deliberately [13, 31, 32]. Additionally, to our knowledge, we are the first to examine SW ability in a scenario where a searcher must multitask between searching and externalizing

knowledge structures, which requires attentional switching. Further investigation is needed to determine the extent to which our results represent generalizable trends.

6 CAVEATS AND LIMITATIONS

Our study involved three experimental decisions that may have impacted our results.

First, during each task, participants were given 40 minutes to gather information and 15 minutes to produce their outline. These time constraints were imposed to keep the first study session under two hours. While time constraints are common in interactive search studies, they can also influence participants' perceptions [19]. It is possible that these time constraints may have influenced our results. For example, low-SW participants reported greater task difficulty with the OrgBox versus Bookmark tool. This effect may have been magnified by the fact that participants had limited time to complete both phases of each task. Future work is needed to better understand the interplay between time constraints, specific cognitive abilities, and specific tools available to searchers.

Second, during the outline generation phase of each task, participants were not allowed to conduct any additional searches. Participants were able to revisit pages associated with passages saved onto the Bookmark or OrgBox tool. However, they were not able to issue new queries to gather additional information. Participants were informed about this constraint before the search phase of the task. This decision was carefully considered. We wanted each phase of the task to involve clear objectives. The search phase involved gathering information and learning about the domain and collection. The outline generation phase involved using the gathered information and acquired knowledge to generate an outline. Other information-seeking studies have used similar "phased" approaches to focus on particular stages of search behavior [37, 41]. However, this is a limitation of our study because it may not resemble how some people might do such a task in everyday life.

Third, our study involved 32 participants who were all university employees (students and non-students). Naturally, our participant pool does not fully represent the broader U.S. population. As described in Section 3.1, we managed to recruit participants from different age groups. However, in other respects, our participant pool was more homogeneous (e.g., 16 reported having a STEM-related degree). Additionally, we did not purposely recruit participants with medical diagnoses such as dyslexia or **attention deficit hyperactivity disorder (ADHD)**, which affect millions of individuals and have been linked to deficits in specific cognitive abilities [40, 69, 73]. In this respect, there are many opportunities for future work. For example, future studies should investigate the influences of knowledge representation tools such as the OrgBox on participants with (versus without) specific medical diagnoses. Additionally, future studies could seek to recruit participants with a *wider* range of psychometric test scores. Doing so might reveal *stronger* (or *different*) effects on the same types of outcomes investigated in our study.

7 CONCLUSION

We have reported on a lab study that investigated the influences of two cognitive abilities—WM and SW ability—during complex search tasks assisted by two different types of tools—the OrgBox knowledge representation tool and the Bookmark tool. Using the OrgBox, participants could save and (re-)organize textual passages into "boxes" and make/edit notes about passages and boxes. Using the Bookmark, participants could save textual passages, but not organize them nor make notes. Our three research questions considered three types of outcome variables: (RQ1) search behaviors, (RQ2) post-task perceptions, and (RQ3) the quality of outlines produced by participants for a hypothetical article. Our results revealed several important trends.

First, our RQ1–RQ3 results suggest that knowledge representation tools such as the OrgBox can provide important benefits. In terms of RQ3, using the OrgBox, participants produced higher-quality outlines (the task's main deliverable). Specifically, using the OrgBox, participants produced

outlines with more cohesive sets of subtopics. We believe that the OrgBox influenced participants to think about the structure of their outlines earlier in the task (during the search phase). Additionally, during the outline generation phase, it allowed participants to directly map the contents in each “box” to subtopics under the same parent topic. In terms of RQ2, participants perceived the OrgBox to help them organize information, plan their searches, and monitor their progress during the task. This suggests that *externalization* can support metacognitive activities such as goal-setting and monitoring during complex search tasks. Finally, in terms of RQ1, the OrgBox influenced participants to search more *deliberately*—to search for information for specific “boxes”.

Second, our RQ1–RQ2 results suggest that tools such as the OrgBox can provide specific benefits for users based on their WM capacity. Specifically, in terms of RQ1, our results suggest that the OrgBox influenced high-WM participants to search even *more* deliberately than low-WM participants. In terms of RQ2, using the OrgBox, low- and high-WM participants reported similar levels of difficulty in determining when they had enough information to complete the task. Conversely, using the Bookmark, low-WM participants reported greater difficulty than high-WM participants.

Third, our RQ2 results suggest that tools such as the OrgBox can provide specific benefits while simultaneously introducing challenges for users based on their SW ability. Specifically, high-SW participants reported greater satisfaction and lower difficulty with the OrgBox. Conversely, low-SW participants reported greater satisfaction and lower difficulty with the Bookmark. Compared to the Bookmark, the OrgBox required *attentional switching* between different physical and cognitive activities. Consequently, engaging with tools such as the OrgBox can be more challenging for low-SW searchers.

In terms of future research directions, our results have four important implications. The first two are methodological implications. The last two are related to open questions for future research to consider.

First, in RQ2, we considered the effects on participants’ post-task perceptions. In addition to perceptions commonly investigated in IIR studies (e.g., difficulty, satisfaction), we also considered the extent to which participants perceived the Bookmark and OrgBox to support different (meta)cognitive activities (e.g., organizing information, drawing inferences, goal setting, monitoring progress, evaluating/revising strategies). Our RQ2 results found that participants perceived the OrgBox to provide greater support for these activities. In the field of education, it is widely accepted that engaging in (meta)cognitive activities such as goal-setting and monitoring leads to better learning outcomes [7, 8, 83]. Thus, future IIR studies should continue to measure the extent to which tools support these types of (meta)cognitive activities. In our study, we focused on participants’ *perceptions* of support. Future studies might also consider the extent to which tools promote such (meta)cognitive activities (e.g., by analyzing concurrent think-aloud comments).

Second, our RQ2 and RQ3 results underscore the importance of considering outcome variables that capture both self-reported and objective measures of task performance. In our study, none of our independent variables (i.e., interface, WM, SW) had a significant effect on participants’ perceptions about the quality of their own outlines (RQ2). However, our qualitative analysis of outlines revealed that participants using the OrgBox produced outlines with more cohesive sets of subtopics. Self-reported perceptions of task performance can be influenced by contextual and individual factors such as the quality of tools available and the individual’s unique abilities. For instance, participants may report that they are satisfied with their performance *given* the limitations of the tools available and/or their own abilities. Participants may adjust performance expectations depending on such factors, which may lead to no differences in self-reported performance measures in cases where there are, in fact, objective differences.

Third, our results suggest that knowledge representation tools such as the OrgBox can provide benefits, but also introduce challenges. Cognitive abilities such as WM and SW may *moderate* the

effects of such tools. Based on our results, the OrgBox had a greater effect on the behaviors of high- versus low-WM participants, influencing high-WM participants to search even more deliberately. Additionally, interacting with the OrgBox was perceived as less challenging and more welcoming for high- versus low-SW participants. Our results suggest that knowledge representation tools such as the OrgBox should not be designed as “one size fits all” solutions. An important question for future research is: How can we design knowledge representation tools to better support low-WM and low-SW searchers?

Finally, prior research has studied the effects cognitive abilities such as WM capacity [3, 31, 32], perceptual speed [9], and attention control [3] on search behaviors and outcomes. To our knowledge, we are the first to study the effects of SW ability, which is related to cognitive flexibility. Future work should continue investigating the role of SW ability during scenarios where a searcher must switch between different types of cognitive activities and perspectives. Prior work has developed a wide variety of assistance tools that help searchers organize and annotate information during complex search tasks [4, 33, 35, 36, 57, 79]. Our results suggest that SW ability may influence a searcher’s willingness and/or ability to effectively engage with such tools based on their design.

Taking a step back, our goal was to investigate the influences of two tools on searchers with different cognitive abilities. Such research is important for several reasons. First, it can inform the design of search environments for specific user populations. Second, it can inform the design of *dynamic* search environments that customize tools and interface features to individuals based on their abilities. Third, understanding how cognitive abilities impact search behaviors can help us interpret behavioral measures more reliably.

A APPENDIX

A.1 Pre-task Questionnaire

- [interest] I am interested in this topic.
- [prior_know] I already know a lot about this topic.
- [diff] I think it will be difficult to complete this task.
- [diff_search] I think it will be difficult to search for information to complete this task.
- [diff_integrate] I think it will be difficult to integrate the information I find to complete this task.
- [diff_enough] I think it will be difficult to decide when I have enough information to complete this task.
- [a priori] Right now, I understand what this task is asking me to do.
- [a priori] Right now, I understand the terminology in the task description.
- [a priori] Right now, I understand what I need to learn.
- [a priori] Right now, I know what information might be useful to complete this task.
- [a priori] Right now, I know what my approach to this task will look like.
- [a priori] Right now, I know where to begin my search.

A.2 Post-task Questionnaire

Post-task questionnaire items about interest/knowledge increase, difficulty, satisfaction, and self-assessed expertise in the task domain:

- [interest] My interest in this topic has increased.
- [knowledge] My knowledge of this topic has increased.
- [sat_amount] I am satisfied with the amount of information I found to complete this task.
- [sat_quality] I am satisfied with the quality of information I found to complete this task.

- [sat_strategy] I am satisfied with the strategy I took to find information for this task.
- [sat_time] I am satisfied with the amount of time I spent on this task.
- [diff_task] It was difficult to complete this task.
- [diff_find_info] It was difficult to search for information to complete this task.
- [diff_integrate] It was difficult to integrate the information I found to complete this task.
- [diff_enough_info] It was difficult to decide when I had enough information to complete this task.
- [self_expertise] I feel I could understand a debate by policy makers about this topic.
- [self_expertise] I feel I could carry on an intelligent conversation about this topic.
- [self_expertise] I feel I could make an effective argument to defend my opinions about this topic.
- [self_expertise] I feel I could successfully answer someone else's questions about the articles I just read.

Post-task questionnaire items about perceptions of outline quality:

- My outline has a clearly defined scope.
- The contents of my outline can be supported by articles I found during my search.
- My outline reflects multiple different viewpoints about the issue.
- My outline makes broad connections.
- My outline synthesizes information from multiple articles.
- My outline covers all the important aspects of the issue.
- My outline demonstrates depth of understanding.
- My outline reflects a balance between breadth and depth.
- My outline is logically organized.
- My outline is well structured.

Post-task questionnaire items about perceptions of the tool being helpful with (meta)cognitive activities related: to (1) planning, (2) connecting, (3) organizing, (4) goal-setting, (5) monitoring, (6) evaluating, and (7) updating.

- (1) [plan] During the task, the [OrgBox|Bookmark] helped me:
 - Decide how to approach the task.
 - Develop a plan for exploring the topic.
 - Decide where to begin my search.
 - Set goals for what information I needed to find.
- (2) [connect] During the task, the [OrgBox|Bookmark] helped me:
 - Improve my understanding of ideas important to the topic.
 - Consider connections between ideas.
 - Consider how parts of the topic relate to each other.
 - Make connections between the information I found and my own prior knowledge.
- (3) [org] During the task, the [OrgBox|Bookmark] helped me:
 - Organize the information that I found.
 - Structure my thoughts.
 - Figure out how the information I found fit together.
 - Understand the structure of the topic related to the task.
- (4) [goal_setting] During the task, the [OrgBox|Bookmark] helped me:
 - Search for specific information.
 - Search broadly across the topic (broaden my search).
 - Search deeply on some aspects of the topic (deepen my search).

- (5) [monitor] During the task, the [OrgBox|Bookmark] helped me:
 - Keep track of my progress toward goals that I set for myself.
 - Decide when a goal was successfully accomplished.
 - Decide when to transition from one goal to the next.
- (6) [evaluate] During the task, the [OrgBox|Bookmark] helped me:
 - Evaluate whether the information I found was useful.
 - Evaluate whether my strategy for the task was working.
 - Evaluate whether I was working fast enough to find all the needed information.
- (7) [update] During the task, the [OrgBox|Bookmark] helped me:
 - Change or revise my approach to the task.
 - Change or revise my understanding of the task itself.
 - Change or revise my feelings about the task.
 - Change or revise how much effort I devoted to a specific goal.
 - Change or revise my level of confidence during the task.

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